

(21) Cyclone

1. AIR MASS , Cyclone



Atmospheric circulation model

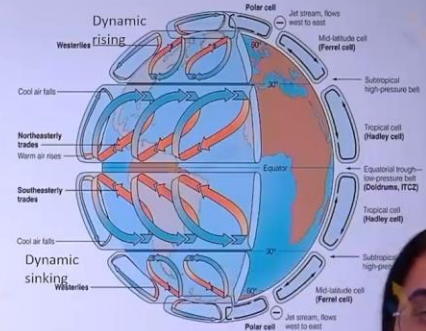
To exit full screen, press Esc

Explained

- Planetary winds: Trade wind, Polar easterlies, Westerlies
- Subpolar rains and cyclone
- Deserts at subtropics: Sahara, Namib, Atacama, California, Arabia, Australia

Ignored

- Upper tropospheric circulations is not correct: Considers upper tropospheric wind as mirror opposite of surface winds. In reality vertical cells do not extend upto tropopause but are shallow (Extends within 1 km)
- Local & regional condition not explained: Monsoon,
- Simplistic Assumption: Land water contrast is reality. In reality there are pressure cells



Tri-cellular
Ferrel's model



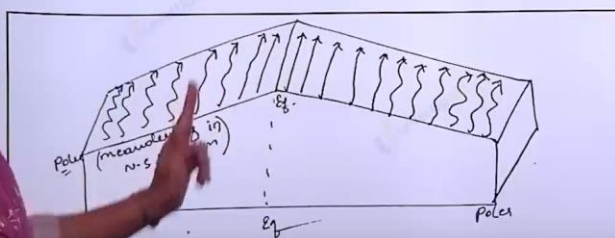
Pressure Cells on Earth considering land water differential

- Latitudes which has same nature across the surface will have pressure belts: ELPB, PHPB (N), PHPB (S), SPLPB (S).
- Pressure Cells rather than belts are seen in SPLP Cells (N), STHP Cells (N and S) and Oceans have permanent cells
- SPLP Cells (North) which are permanent: Aleutian LP Cell, Icelandic LP Cell
- STHP Cells (South) which are permanent: Mascarene HP, Pacific HP, Atlantic HP
- STHP Cells (North) which are permanent: Hawaii High and Azore High

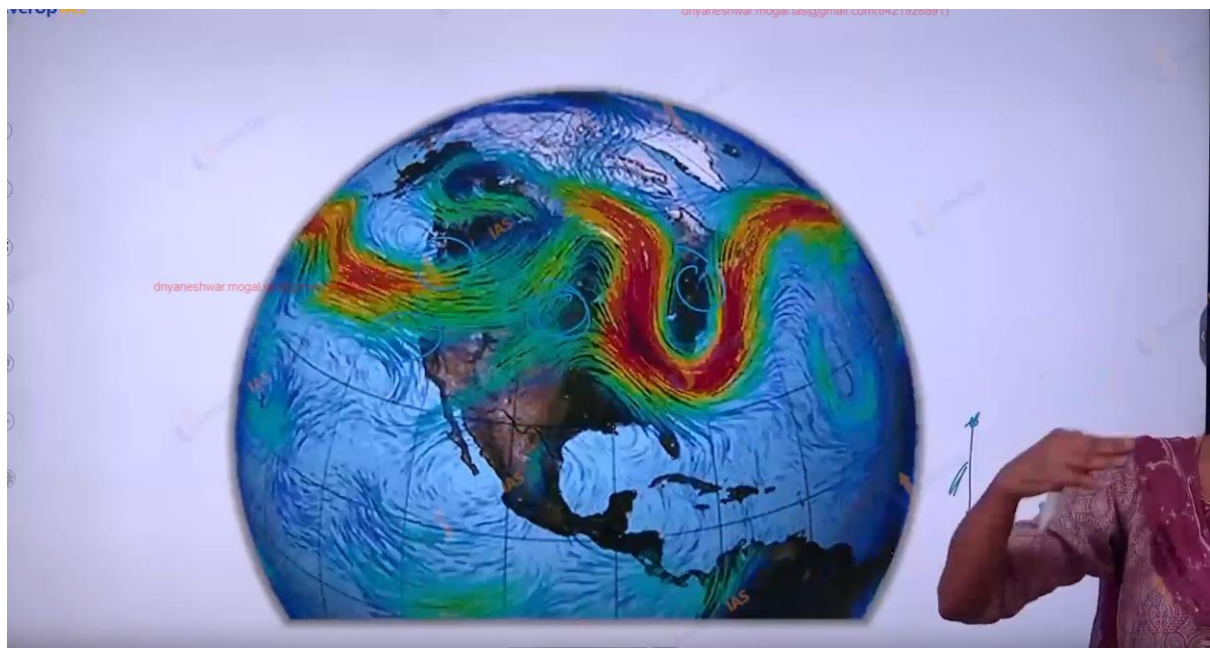
Rossby Wave

- These are upper tropospheric winds.
- Move from west to east.
- Meandering increases as we move from Equator to poles
- Meandering becomes very sharp beyond 55 degree N/S

Rossby Wave



<https://oceanservice.noaa.gov/facts/rossby-wave.html>





Jet Stream

<https://scijinks.gov/jet-stream/>

- Velocity of Jet stream increases because of **temperature difference** between the cells which provides an **additional force** which increases speed of Jet stream.
- **Some Important Jet Stream are:** Sub Tropical Westerly Jet Stream, Polar Jet Stream, Polar Night Jet Stream, Tropical Easterly Jet Stream.

The diagram shows the Earth with the Prevailing Jetstream highlighted in blue. A large red 'H' is placed over the Arctic region, indicating a high-pressure system. The text 'PREVAILING JETSTREAM' and 'IS ENHANCED BY ARCTIC AMPLIFICATION' is displayed. The source 'clideo.com' is mentioned in the bottom right corner.

Rossby Wave

- Meandering Rossby wave create additional pressure pattern in closed loop cells of alternate High Pressure and Low Pressure in the upper troposphere
- The pressure pattern on upper troposphere creates additional pressure pattern on Earth's surface
- Rossby wave can go through a life cycle of 10-15 days where they change from near latitudinal west to east straight line to meandering north to South Path: INDEX CYCLE OF ROSSBY WAVES.

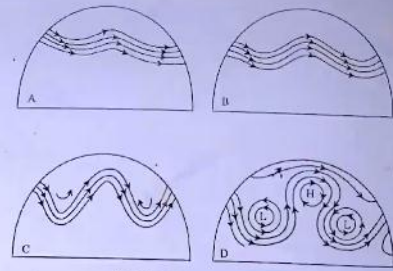
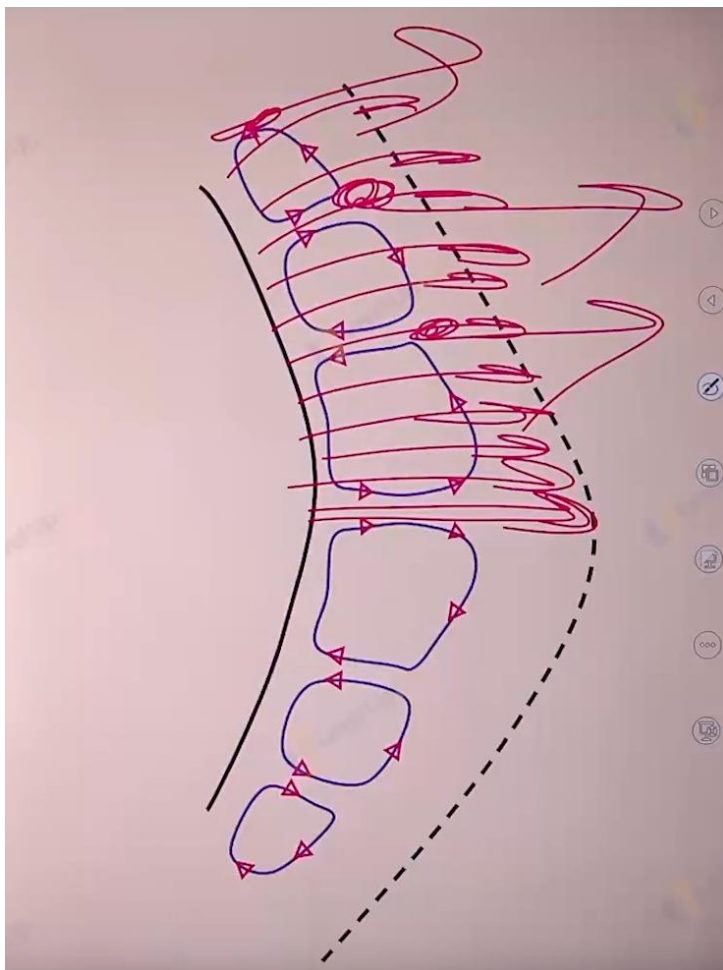
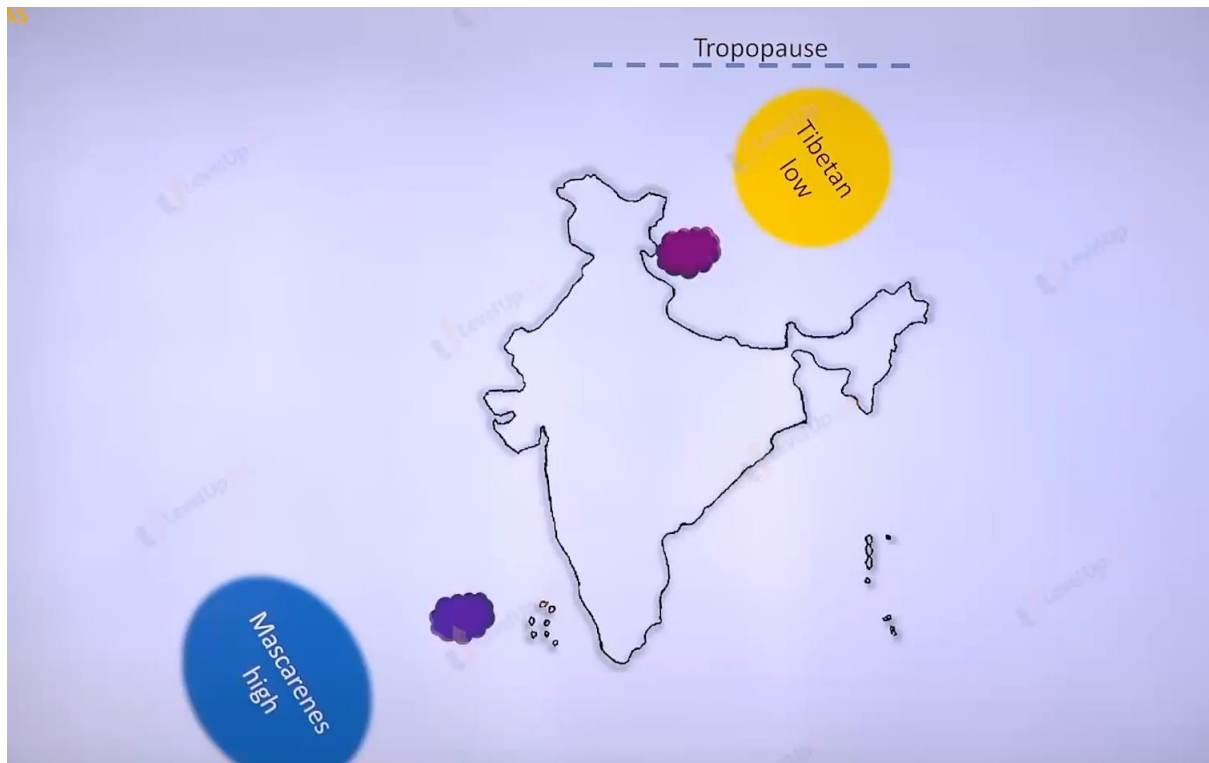


Fig. 6.10 : Index cycle of jet streams.

Dnyaneshwar mogal

dnyaneshwar.mogal.ias@gmail.com



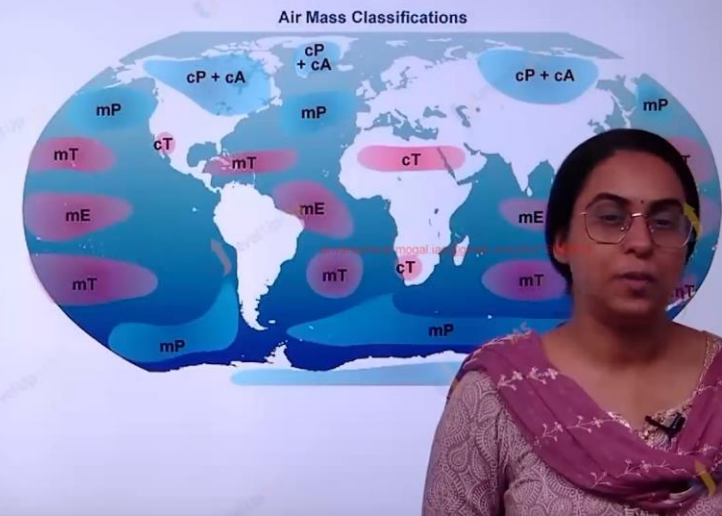


Cyclone: low pressure cell where the air circulates. There are 2 types of cyclones

1. Tropical cyclone: tropic 30 to 10 degree north and 30to 10 degree south
 2. Temperate cyclone: temperate cyclone 35 to 55 degrees north and 35 to 55 degrees south.
-

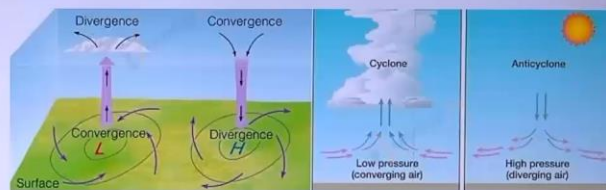
Air Mass

- These are massive bodies of air covering thousands of sq km (horizontally across the Earth's surface) and for entire height of troposphere.
- Airmass have nearly same temperature and moisture levels horizontally.
- Nature and properties of airmasses are acquired from the surface over which airmass develops.



Air Mass Formation: Properties of Source Regions

- Air masses form over source regions that give them their uniform temperature and humidity. **They form on relatively flat surface and extensive surface**
- **Airmass develop in region of High Pressure Conditions and atmospheric stability.** These condition prevent these large volumes from moving. So, while they stay stationary over a region, they acquire the conditions of that region, either temperature or humidity: **Siberian High Pressure condition in winters, Canadian High Pressure Conditions in Winters.**



Types of Air Masses

Based on places of the formation: There can be 4 types of air masses:

- **Antarctica:** These air masses form in the Antarctica region and are very cold.
- **Arctic:** These air masses form in the Arctic region and are very cold.
- **Tropical:** These air masses form in low lying latitudes and are warm up to a moderate level.
- **Polar:** These air masses form in the high-latitude region and are cold.

Based on nature of surface: There can be 2 types of air masses:

- **Maritime:** Maritime ones form over the water bodies and are filled with moisture.
- **Continental:** Whereas, the Continental ones form over the land and are arid.
- **Eg:** An air mass formed over Siberia is a continental polar air mass that is cold and dry. However, one that is formed over the Indian Ocean is a tropical air mass that is humid and warm.

Accordingly, following types of airmasses are recognised:

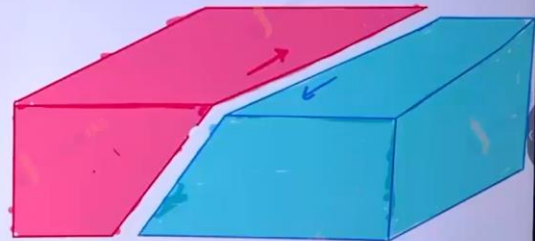
1. Maritime tropical (mT);
2. Continental tropical (cT);
3. Maritime polar (mP);
4. Continental polar (cP);
5. Continental arctic (cA).
6. Continental Antarctica (CAA)

Tropical air masses are warm
polar air masses are cold.

Two most imp airmasses are: cP, mT

Fronts

- It is an inclined zone along which two opposing air masses (having contrasting characteristics in terms of air temperature, humidity, density) converge together.
- The frontal zone is inclined at a low angle to the Earth's surface.
- Along a front, one airmass will be relatively warmer than the other airmass. Warmer Airmass will try to rise along the front and cooler airmass will drag beneath.



CF	WF
- Steep	- gentle
- heavy R.F	- gentle R.F
- R.F of short duration	- R.F of longer duration

Cold Front

- This is formed when cold air mass initiates the movement and hits the warm air mass and makes it rise.
- It is a steep frontal surface and so condensation takes place rapidly.
- Cirrus, Alto stratus, Nimbostratus and Cumulonimbus clouds are formed and heavy rainfall with lightning and thundering takes place.

Cloud development because of frontal lifting of warm moist air

Advancing cold air behind cold front

Receding warm air ahead of cold front

Direction of frontal movement

Cold front map symbol

Cold Front

Cirrus

Altostratus

Nimbostratus

Cumulonimbus

Cold Air

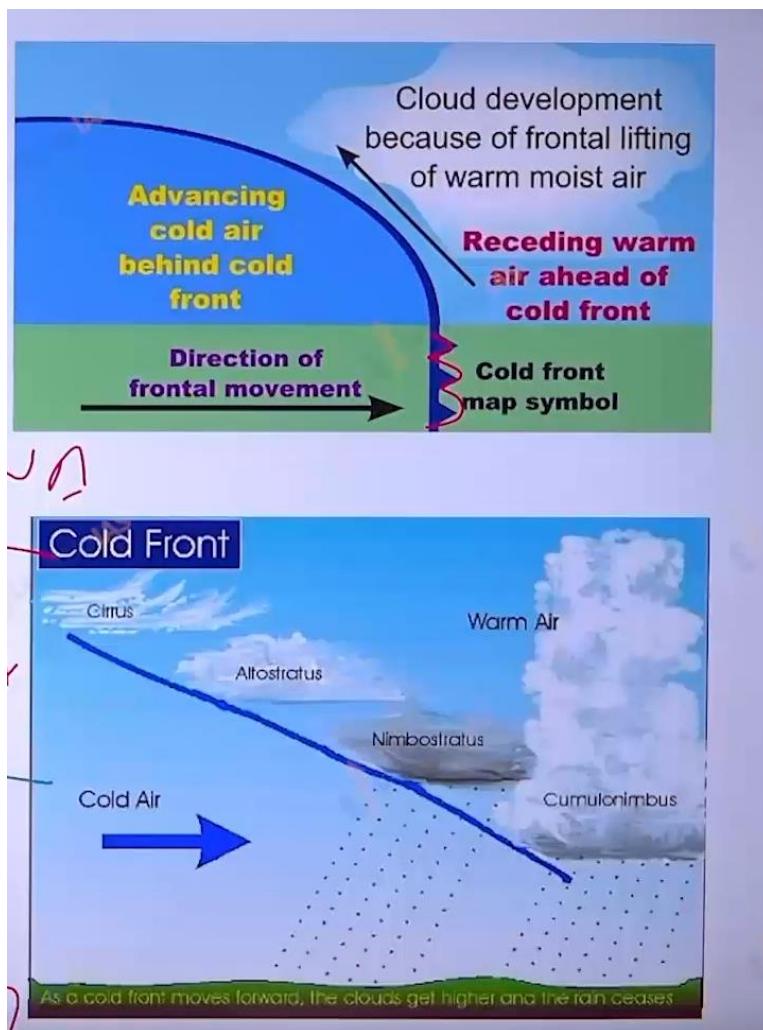
Warm Air

precipitation

e

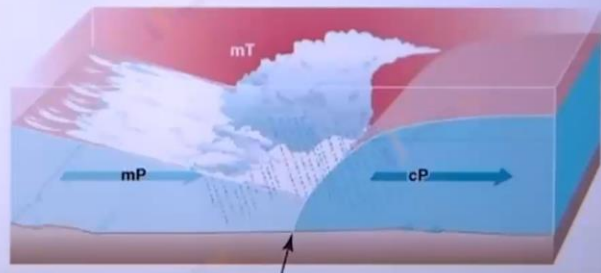
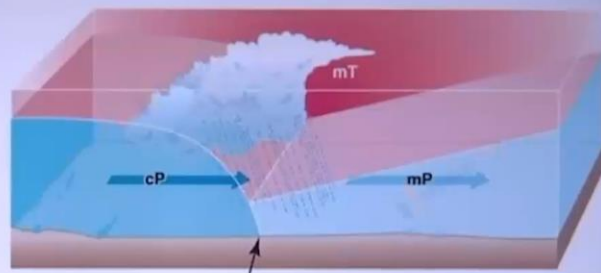
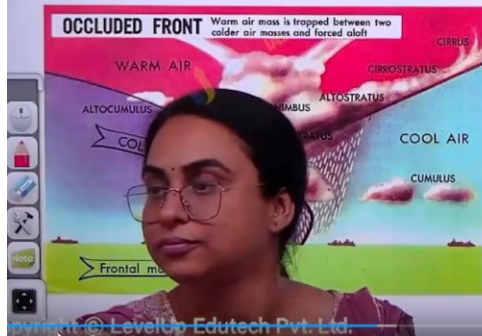
Cold front

1. This is formed when the cold air mass is active, and it moved into the territory of warm air mass
 2. It is steep front and therefore the air raise swiftly passed and the condensation take place rapidly with heavy rainfall of short duration
 3. Cirrus, alto stratus, nimbostratus and cumulonimbus cloud are formed. It leads to heavy rain fall with lightning and thunder taking place.
-



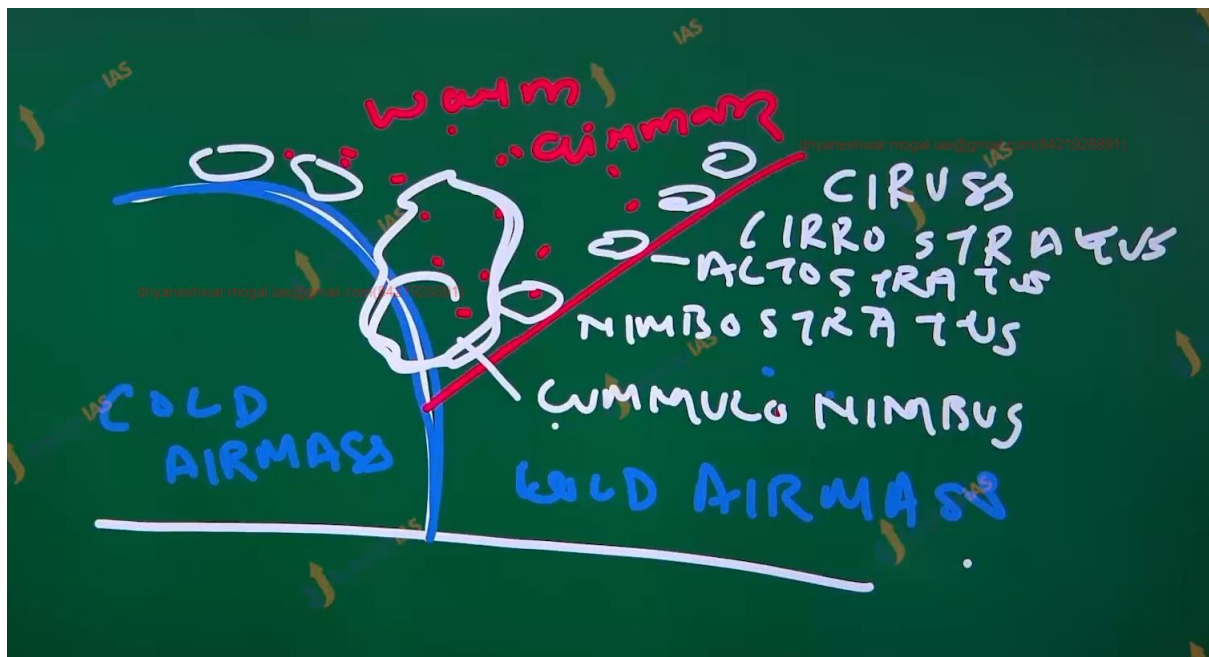
Occluded Front

- When warm air mass is completely raised over the cold air mass it is called an occluded front.
- Weather along this sector is complex—a mixture of cold front type and warm front type weather. These are common in western Europe.



Occluded Front :

1. Warm air complete tossed above the cold air mass
2. Weather along this front is very complex as there is mixture of cold front and warm front.



LevelUp IAS

Air Mass Classifications

©The COMET Program

Department of Atmospheric Sciences
University of Illinois at Urbana-Champaign

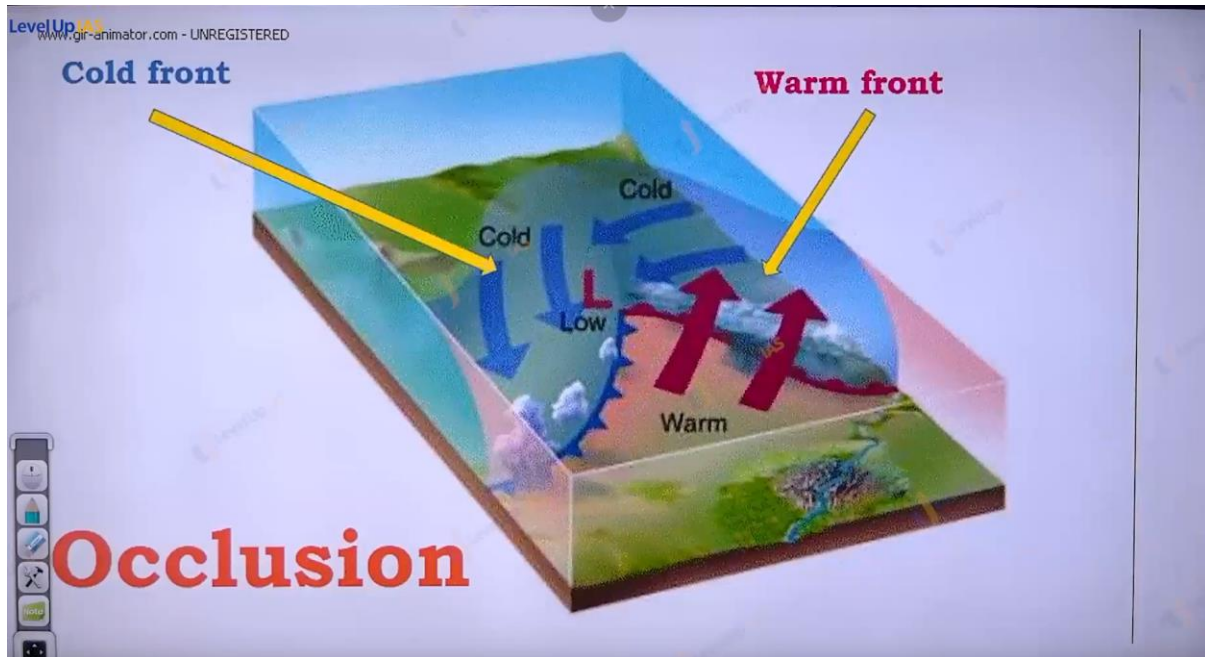
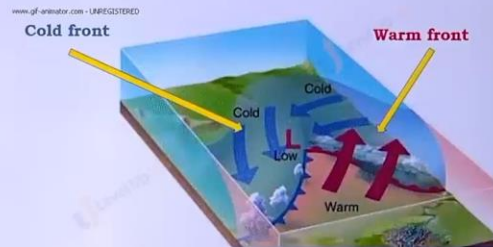
LevelUp IAS
www.gir-animator.com - UNREGISTERED

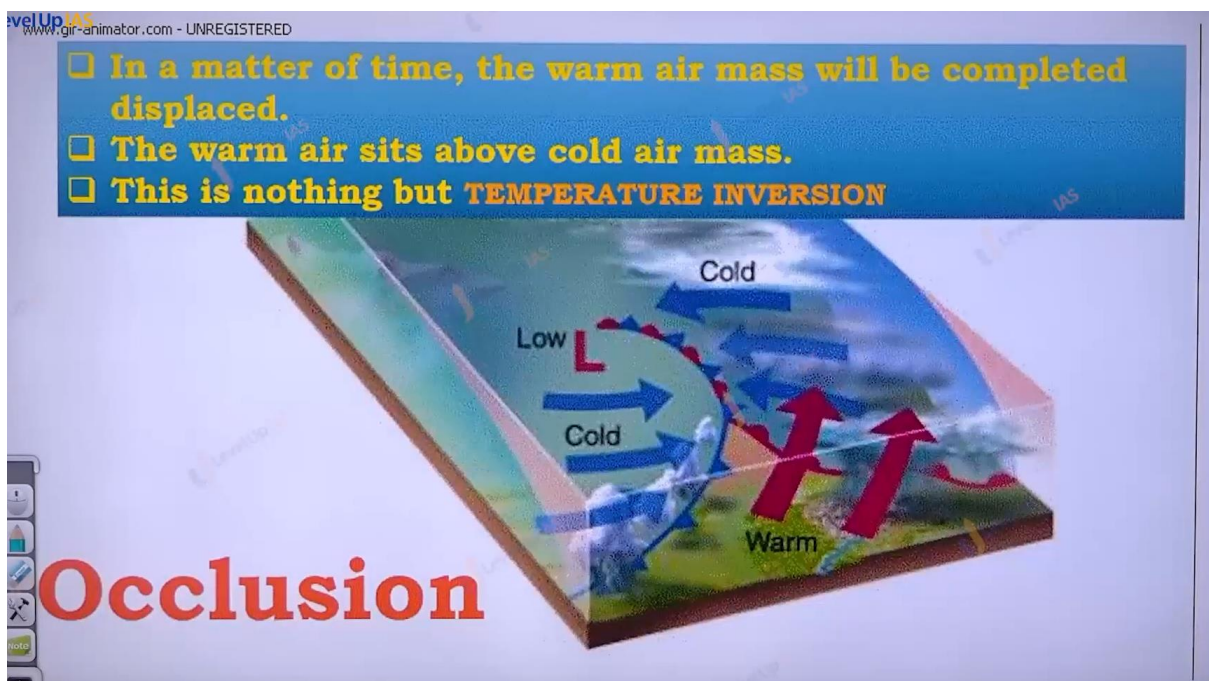
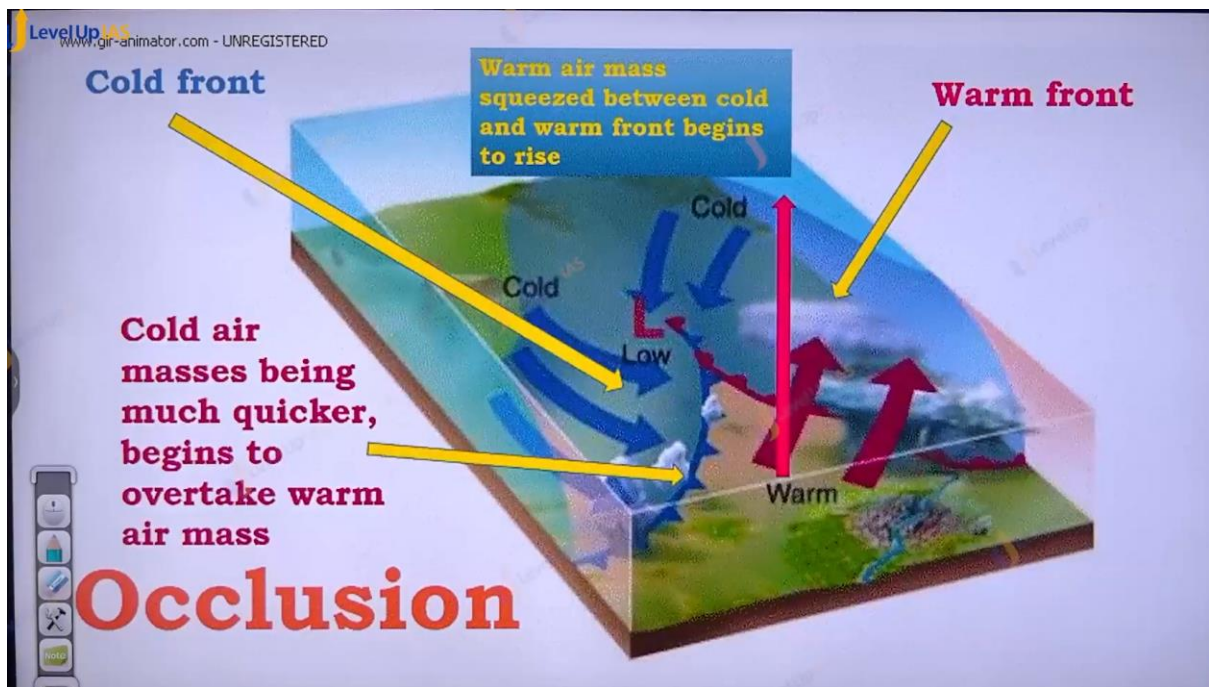
Northern Hemisphere			
1	2	3	4
Southern Hemisphere			
1	2	3	4

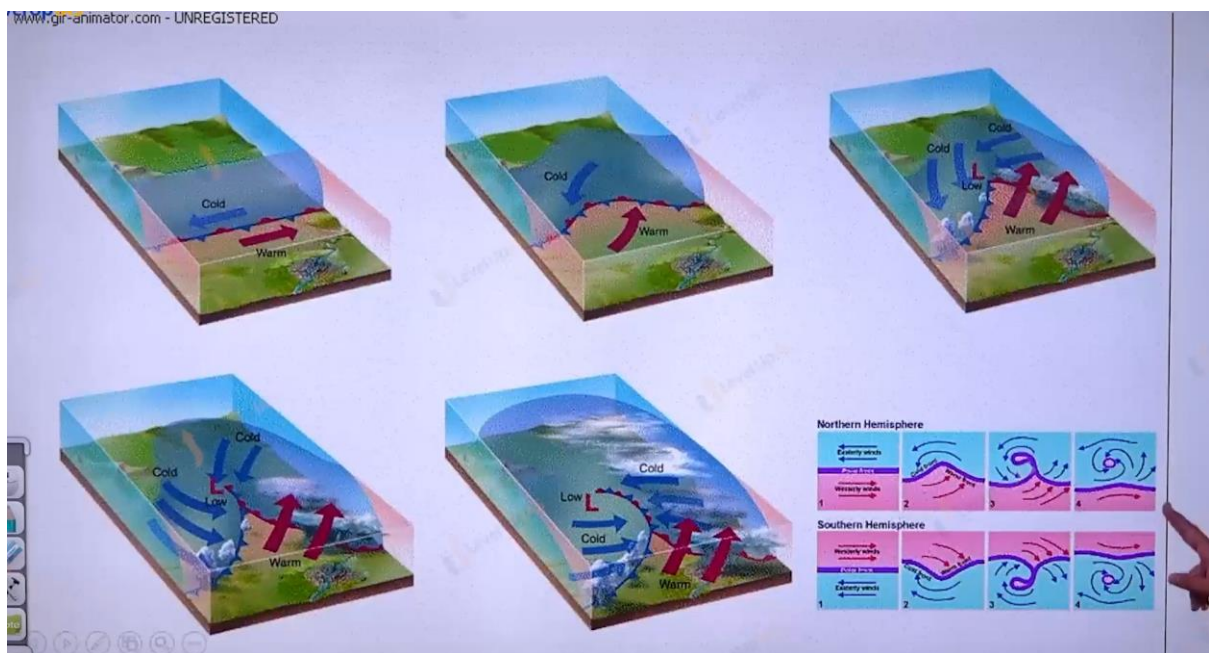
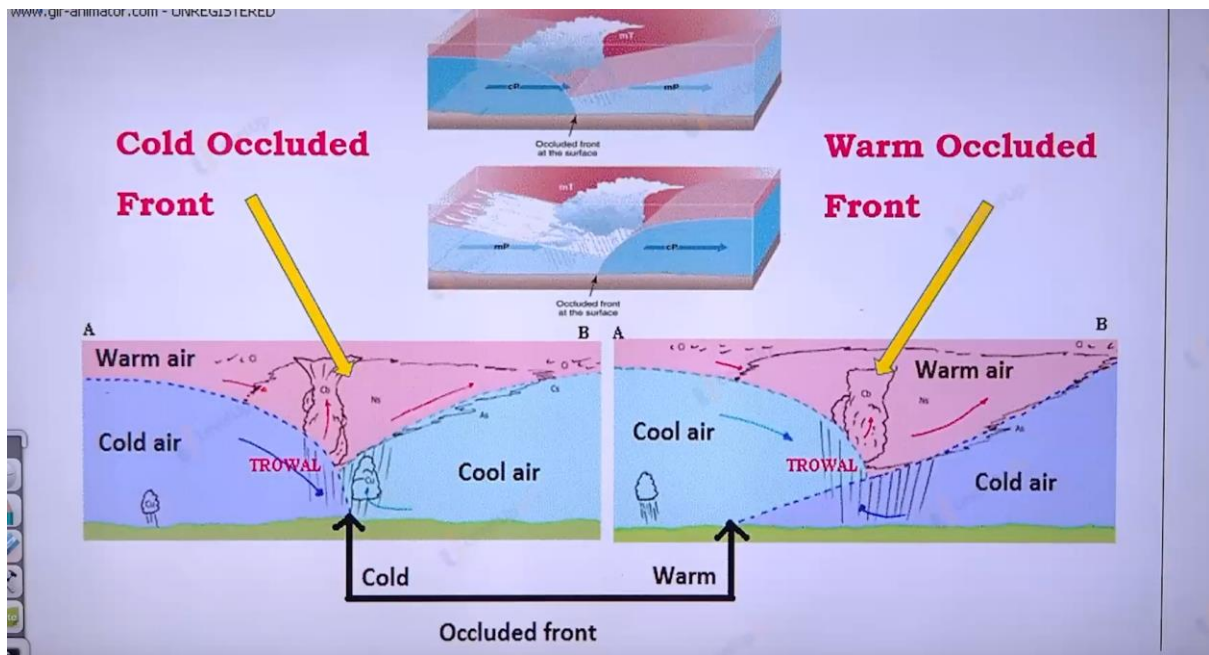
Formation of Temperate Cyclone (Extra-Tropical Cyclones)

- There are pockets of warm air compressed between the forward and the rear cold air.
- The cold front moves faster than the warm front eventually surpassing the warm front.

The warm air is entirely lifted and the front is occluded and the

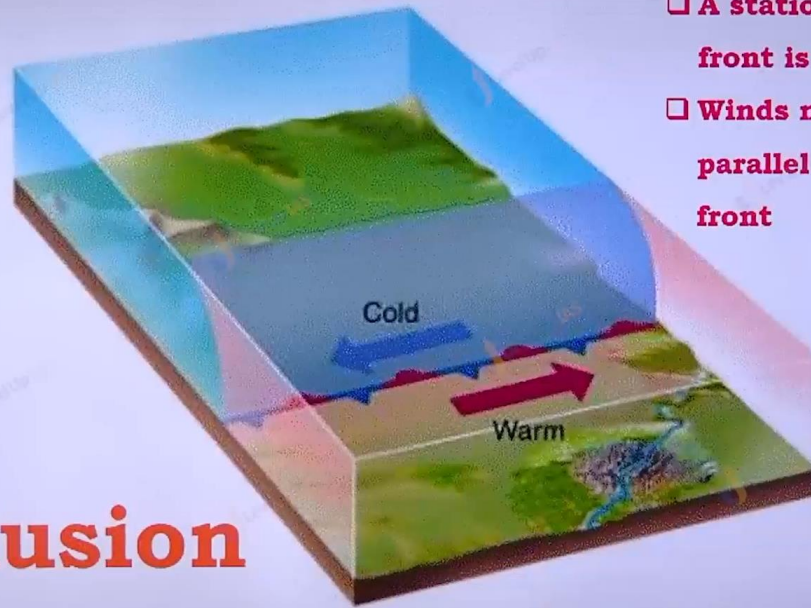






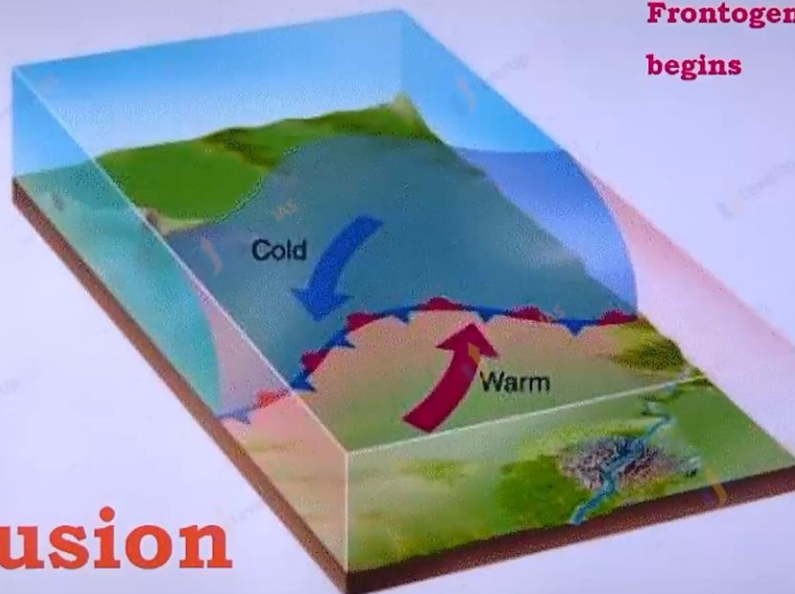


- ❑ A stationary front is created
- ❑ Winds move parallel to the front

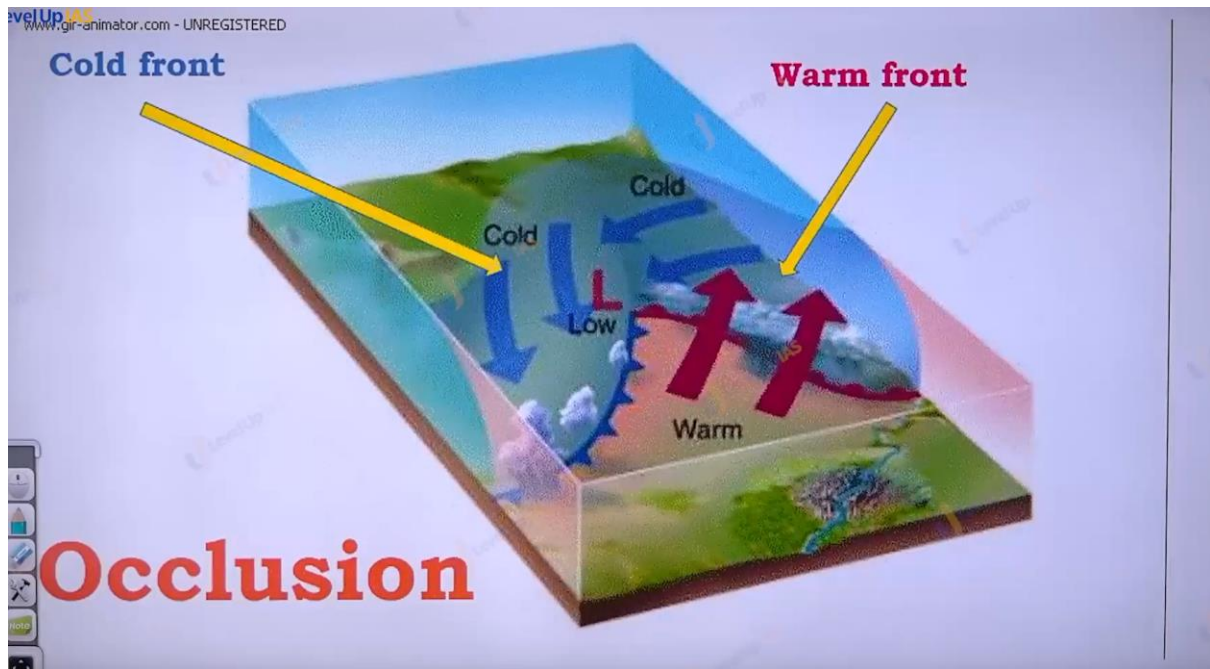


Occlusion

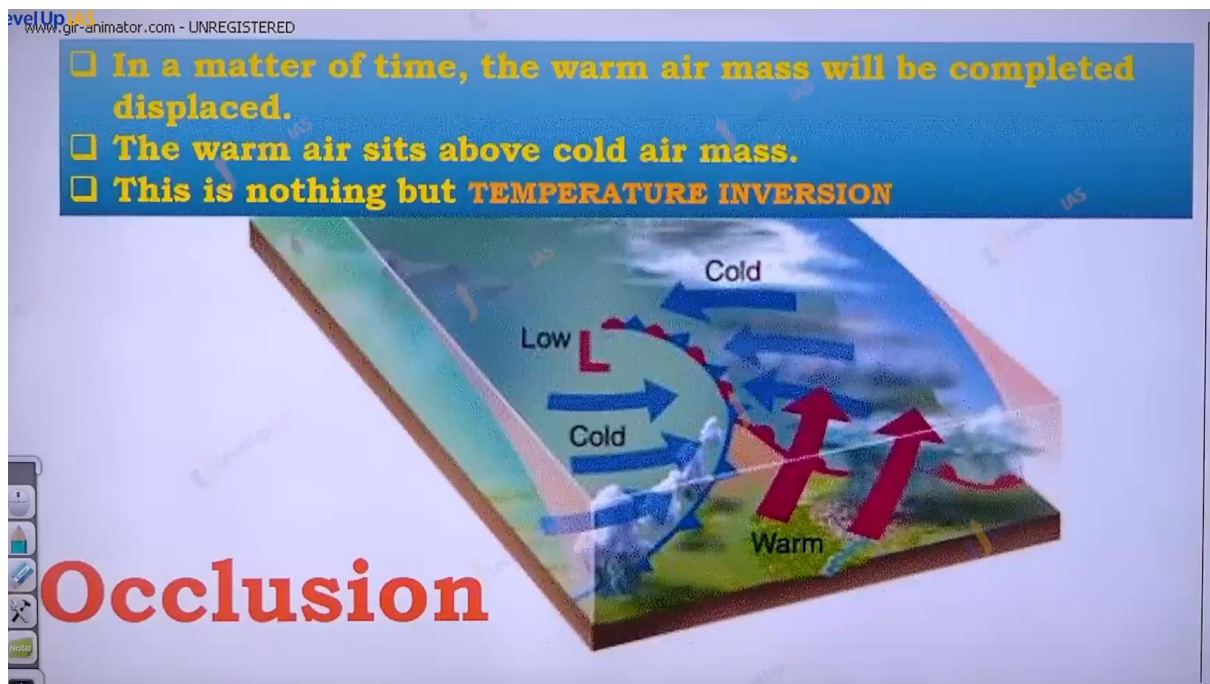
Frontogenesis begins

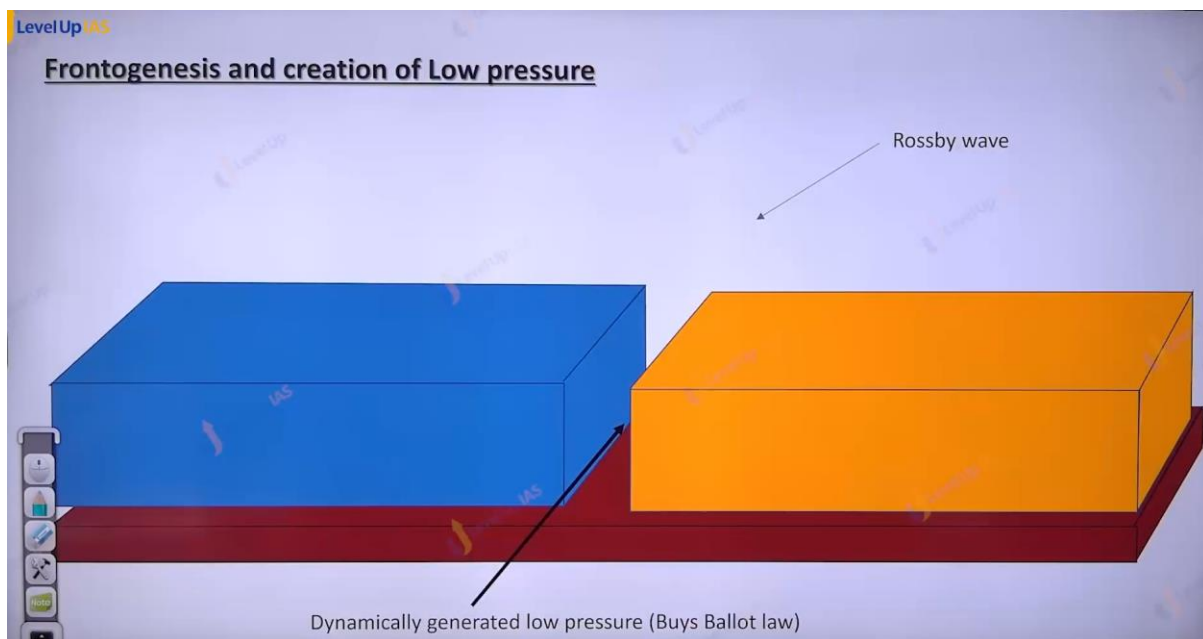
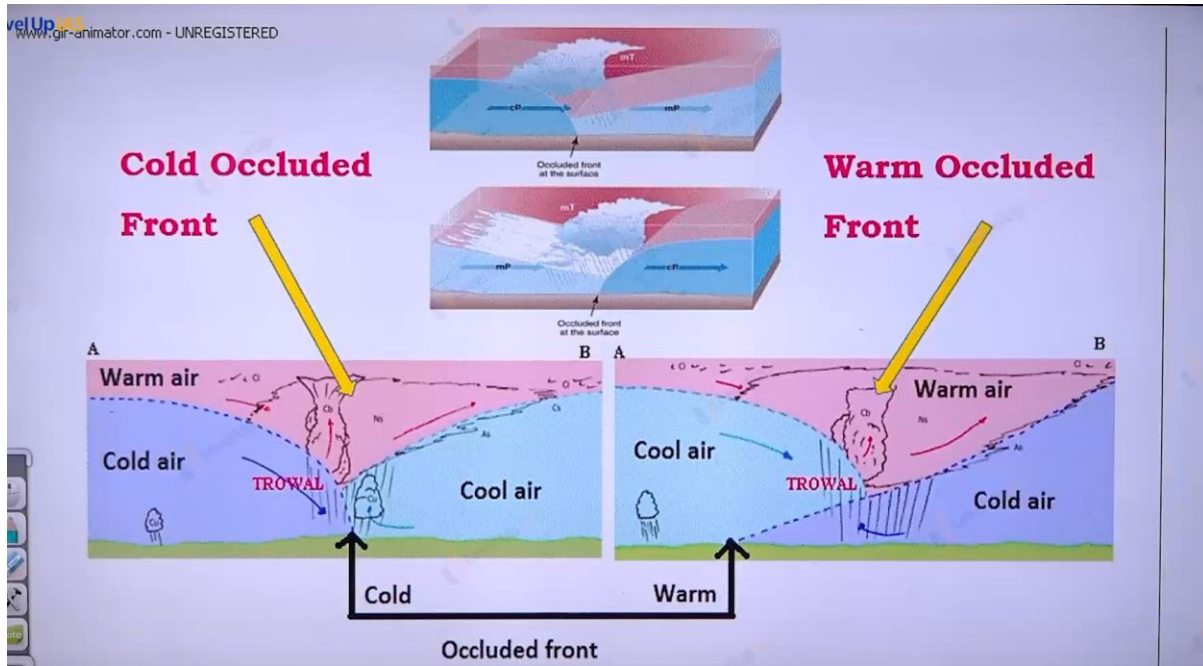


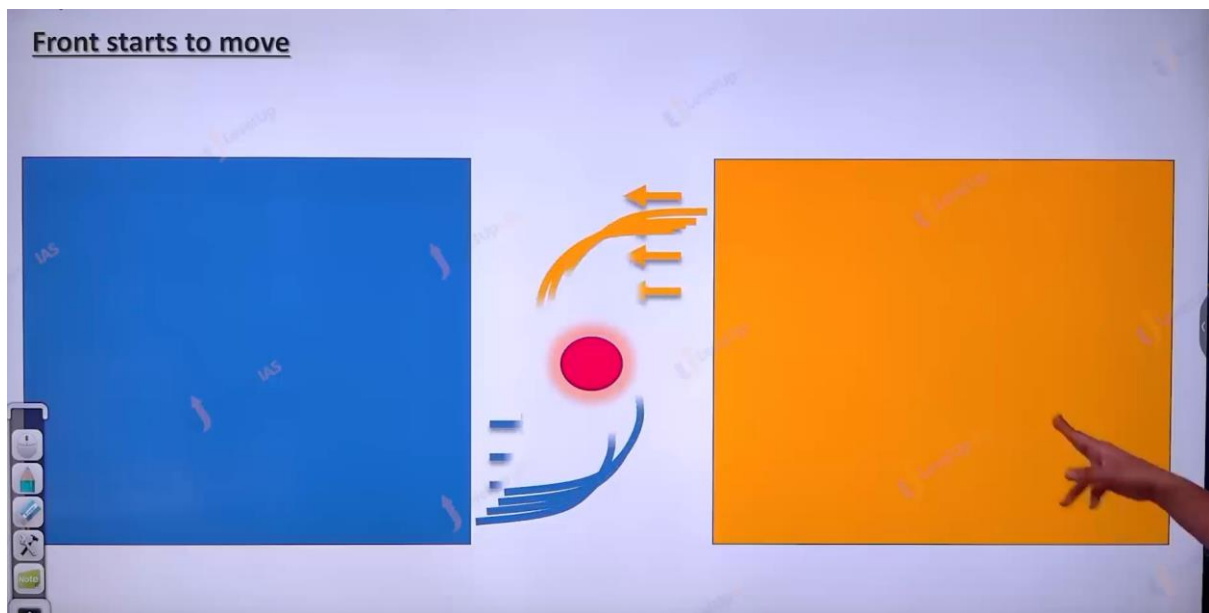
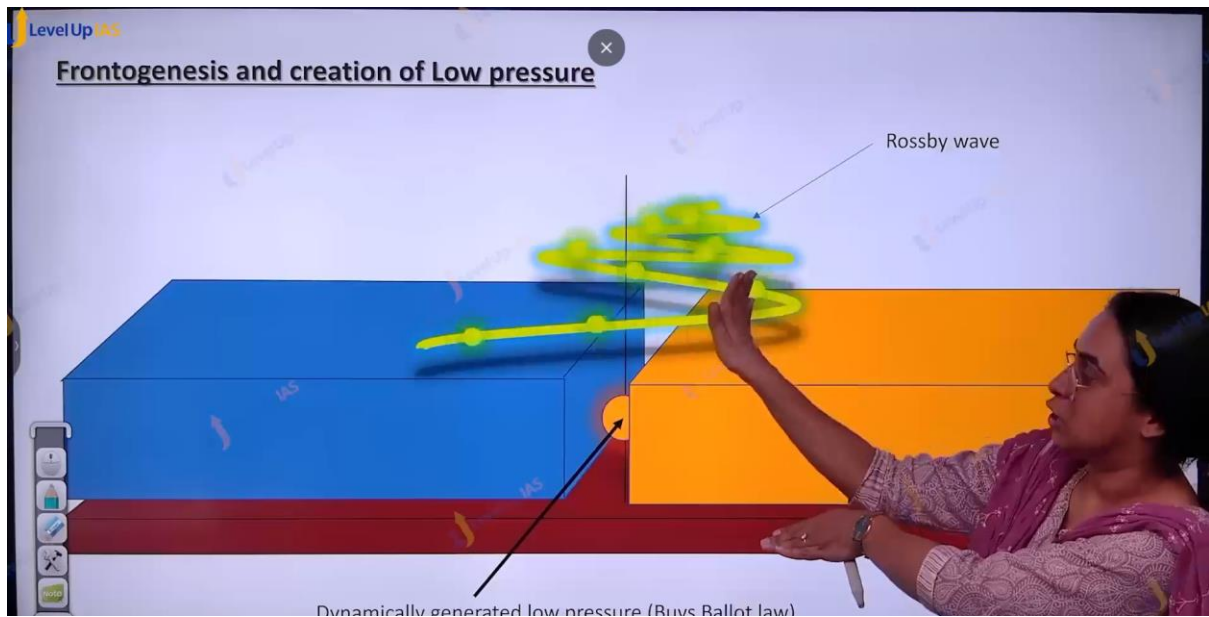
Occlusion

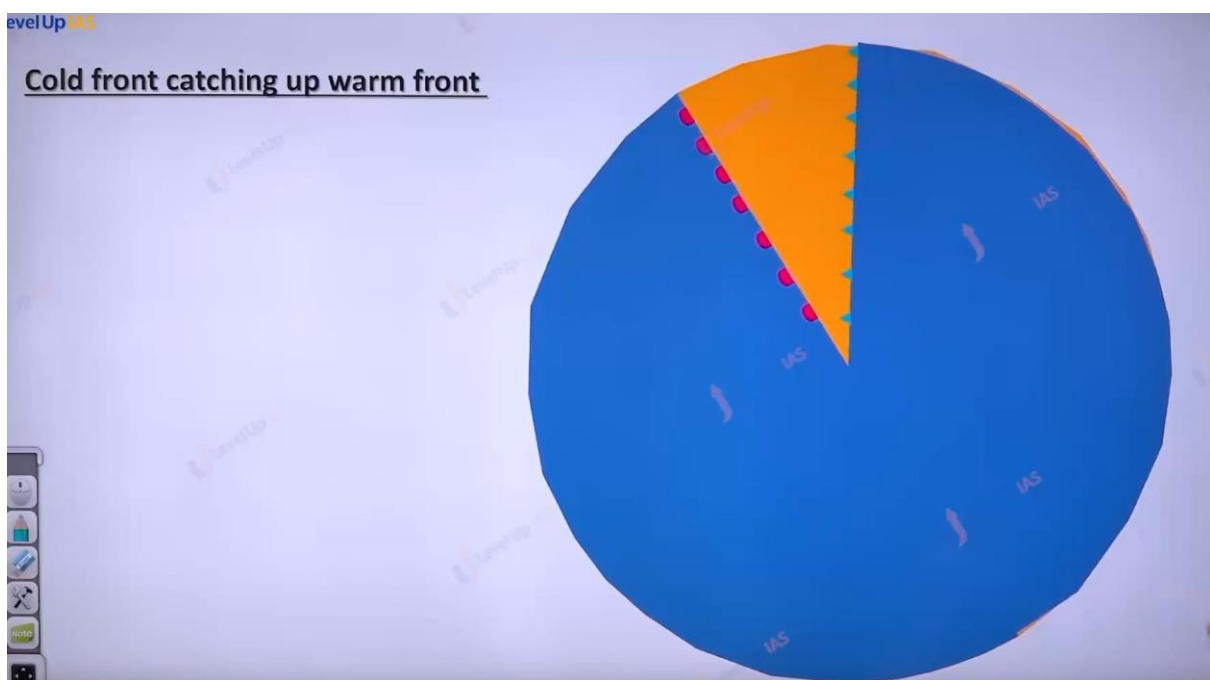
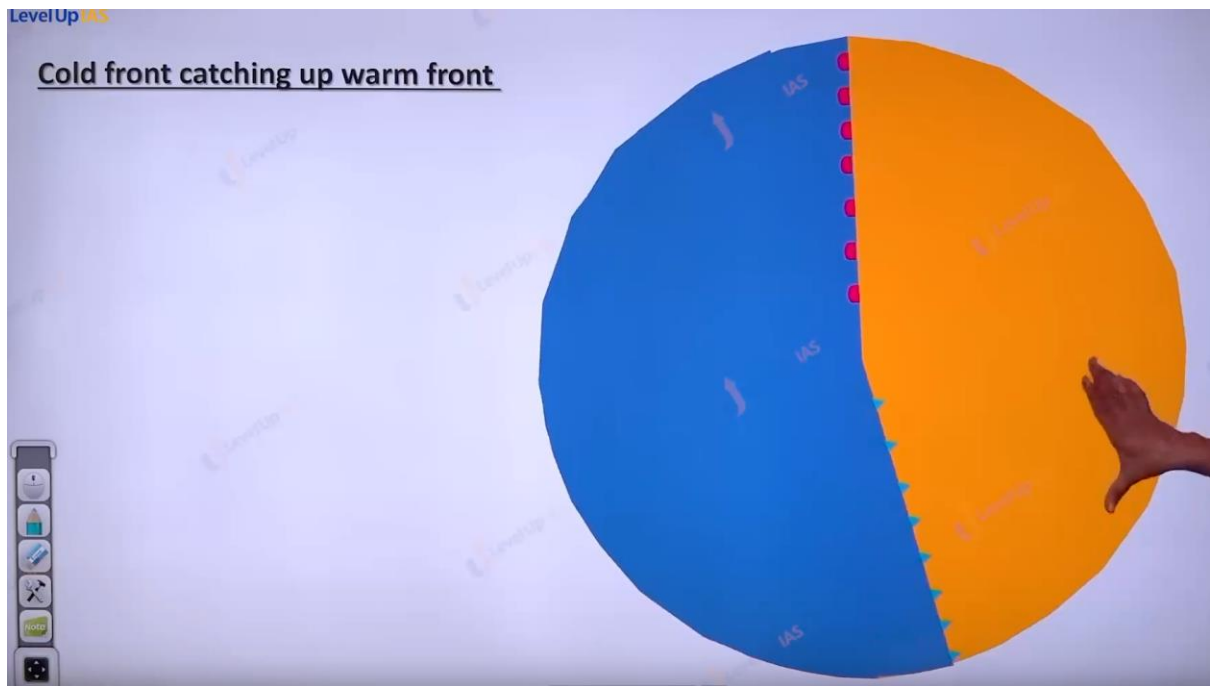


Two front formed, cold is faster therefore the it will reached at the warm front.







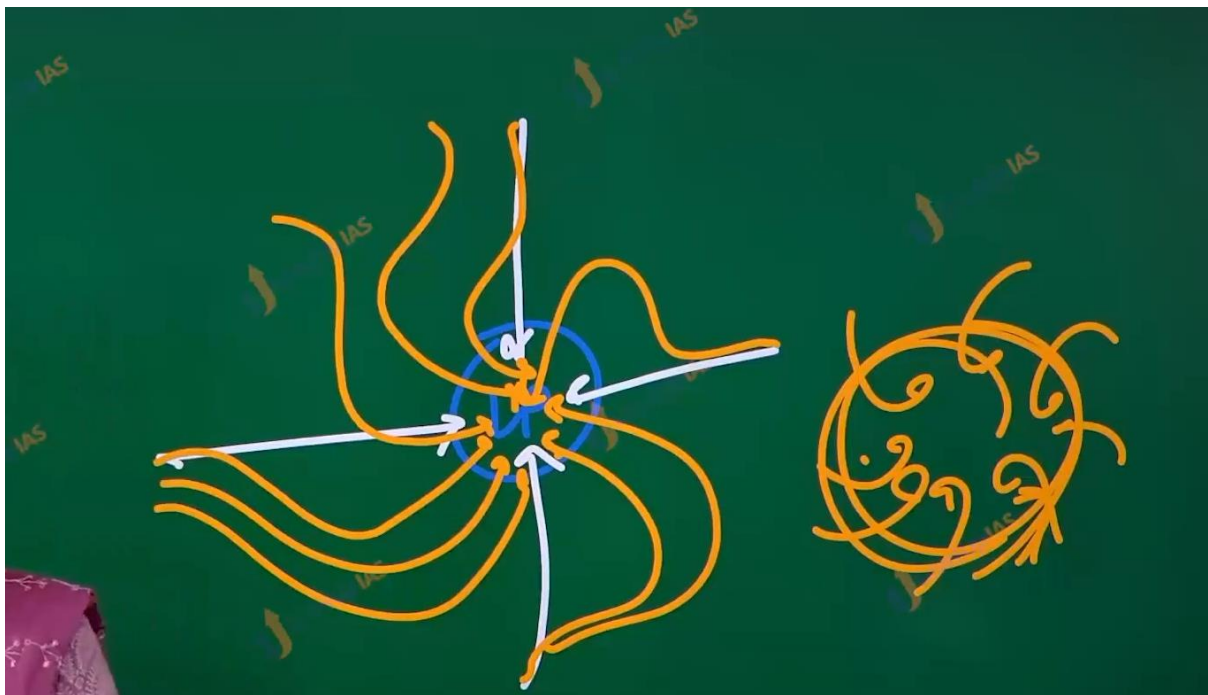


Here yellow is warm air mass and blue is cold air mass . cold air mass is faster it will catch up to warm air mass.

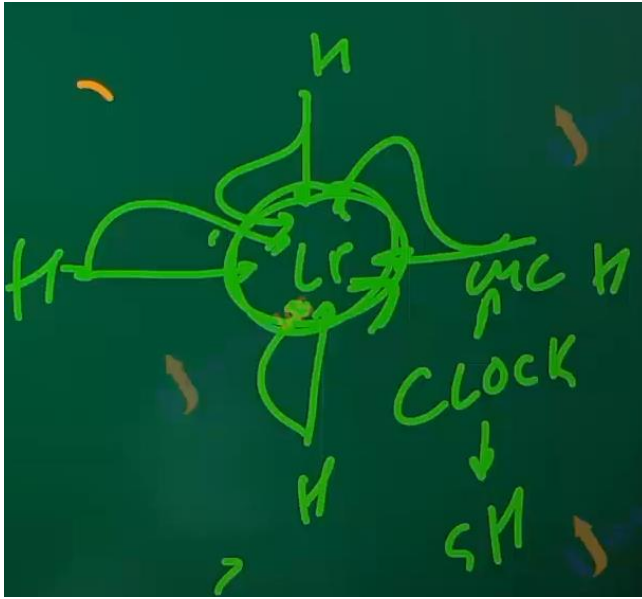
Cyclone

- Are Low pressure System
- Are Rotating System
- Localised low pressure centre with winds converging from all sides.
- Moves Anticlockwise in NH and Clockwise in SH
- At the equator, the Coriolis force is zero so low pressure gets filled instead of getting intensified. That is the reason why tropical cyclones are not formed near the equator.
- Genesis of Low Pressure can be
 - Thermal Origin: Heating and Rising Air. This is seen in tropical latitude: **TROPICAL CYCLONE**
 - Dynamic Origin: Related to Upper Tropospheric Conditions (Rossby Wave) or due to movement of air mass: **TEMPERATE CYCLONE**

(Note: The video also includes satellite images of cyclones in the Northern and Southern Hemispheres, showing their characteristic spiral patterns.)



(this is dirgram of cyclone in the northern hemisphere)



This cyclone in the southern hemisphere.

Cyclone:

-
1. Cyclones are low pressure systems and
 2. They are rotating systems.
 3. Where in cyclone localized low pressure exists with winds converging from all the sides.
 4. They move anticlockwise in the northern hemisphere (cyclone-anticlockwise in northern hemisphere).
 5. At the equator the Coriolis force is zero; therefore, the low pressure is filled instead of getting intensified. This is the reason cyclones do not form at the equator.
 6. The genesis of low pressure can be due to
 - a. thermal nature heating and raising of air; this is seen in the tropical latitude. (Tropical cyclone.) They are warm core cyclones
-

- b. dynamic origin – related to the upper tropospheric winds.
 - c. That is Rossby wave. Also related to the movement of air mass- seen in the temperate cyclone. They are Cold core cyclone.
-

Temperate Cyclone (Extra-Tropical Cyclones)

- Extra-Tropical Cyclones are storm systems emerging in the mid and high latitudes (60 degree – 65 degree), away from the tropics.
- Very extensive ~500km.
- Generates around the dynamically induced low pressure i.e due to the movement of the Upper Tropospheric Wind and movement of front.

In the beginning front is stationary.

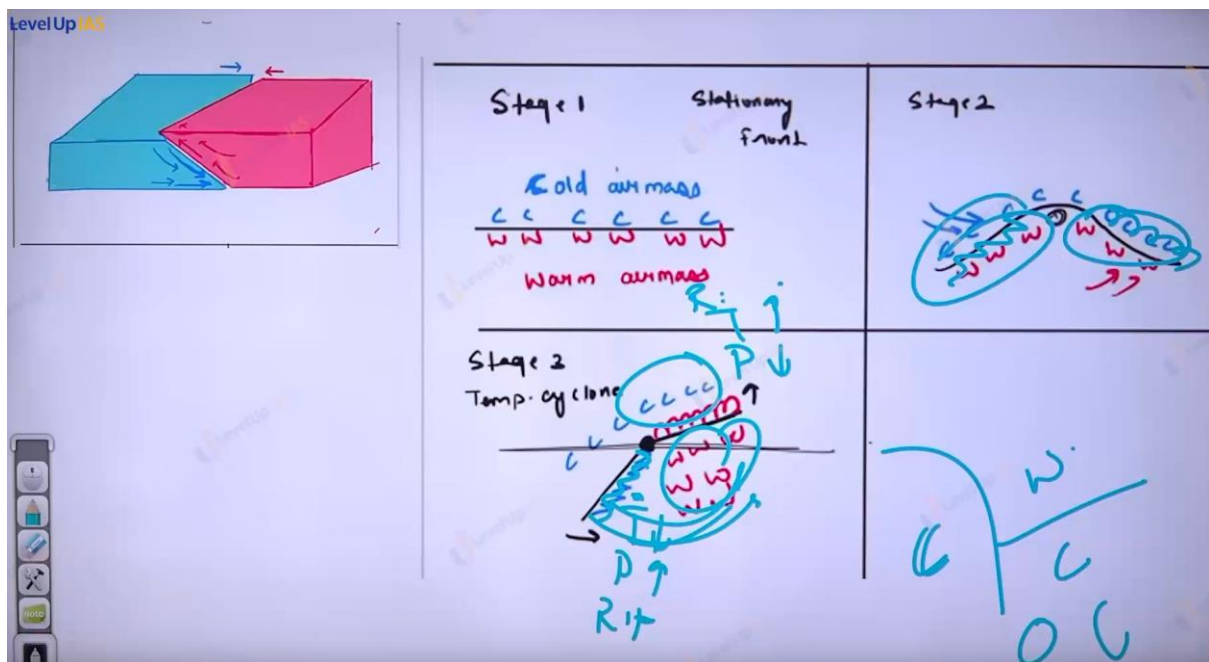
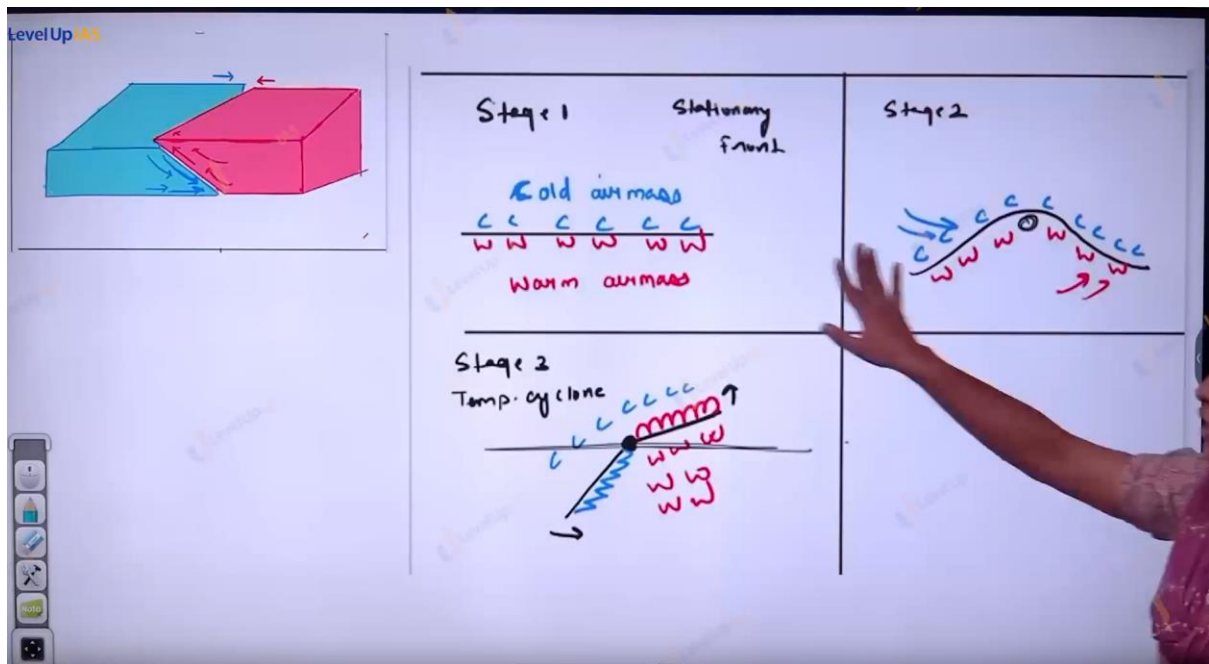


Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gmail.com

Copyright © LevelUp Edutech Pvt. Ltd.

Temperate Cyclone (Extra-tropical cyclones)

1. They are storm system emerging in the middle latitudes 60 – 65 degree latitudes.
 2. They are extensive system.
 3. The temperate cyclone developed around the dynamic induced low pressure that is due to movement of Rossby wave and due to movement of fronts.
-



1. In the beginning the front is stationary then low pressure is developed in the northern hemisphere.
2. The cold air mass blows from the north from the front and warm air mass blows from the south.
3. When the pressure falls along the fronts cold air moves towards the south and warm air moves towards the north setting in motion an anti-clockwise circulation.

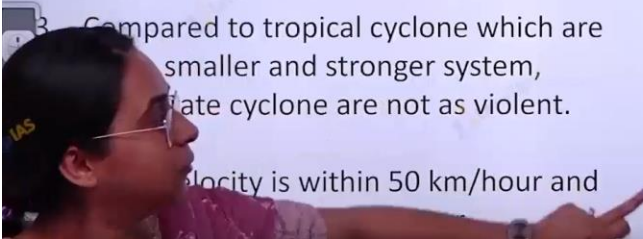
4. cyclonic circulation developed into temperate cyclone cold front and warm front.
 5. There are pockets of warm air compressed between the forward and rear cold air
 6. Cold front moves faster than the warm front, eventually surpassing the warm front.
 7. The entire warm air is entirely lifted and the front is occluded this marks end of temperate cyclone.
-

Features of temperate cyclones:

-
1. They are a massive system which extends 1000 km with the radius exceeding 200 to 300 km.
 2. It is more intense in the winter because of stronger air mass contrast.
 3. Compared to tropical cyclones which are much smaller and stronger system temperate cyclones are not as violent.
 4. The winds velocity is 50 km/hour 80 km/h.
 5. If temperate cyclones move from the west to east under the influence of westerlies.
 6. Temperate cyclones can be developed both land and ocean unlike tropical cyclones that can formed only on the warm water and never on the land.
 7. In the mid latitudes the favorable location for then temperate cyclone permanent low-pressure cell in the sub polar region ex loosen low pressure cell, islanding low pressure cell. And also, along the Mediterranean low-pressure cell.
-

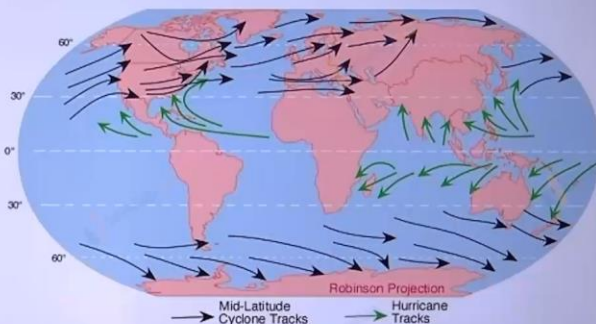
Characteristics of temperate cyclone

1. Temperate cyclone are massive system over thousands of square kilometre with their radius exceeding 200 to 300 km
2. They are more intense in the winters because of the stronger air mass contrast
3. Compared to tropical cyclone which are smaller and stronger system, temperate cyclone are not as violent.
4. Their velocity is within 50 km/hour and



Characteristics of temperate cyclone

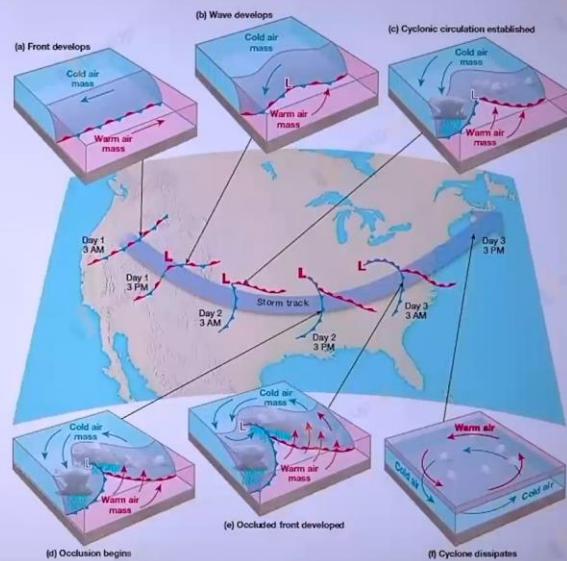
5. Temperate cyclone move from West to east due to the influence of the westerlies
6. Temperate cyclone can form on land and also on waters but they are relatively more developed on the land unlike tropical cyclone that always form on the warm waters and never on land
7. In high mid latitudes following are the preferred location for low pressure development like Aleutian low pressure region, Icelandic low pressure region, Mediterranean Sea low pressure system.



REVISION

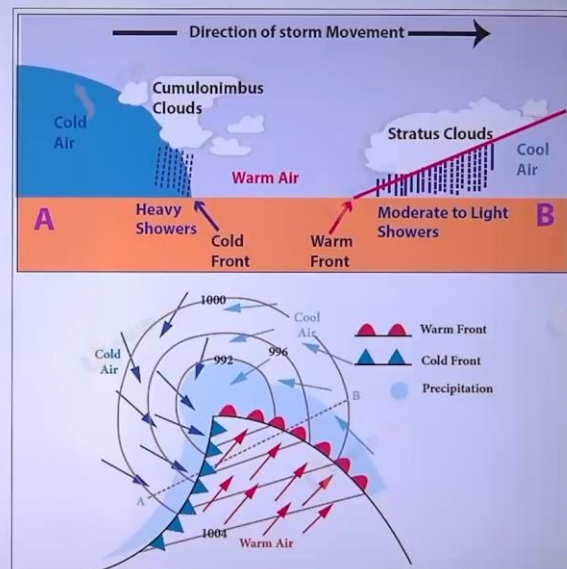
Life cycle of temperate cyclone

- **Stage 1** – low pressure develops between the two fronts
- **Stage 2** – low pressure intensifies, the fronts starts rotating
- **Stage 3** – As the cold air mass is denser it keep tossing the warm air up as the cold front advances.
- **Stage 4** – now cold front catches the warm front and completely occludes it above the ground
- **Stage 5** – the low pressure dissipates gradually.



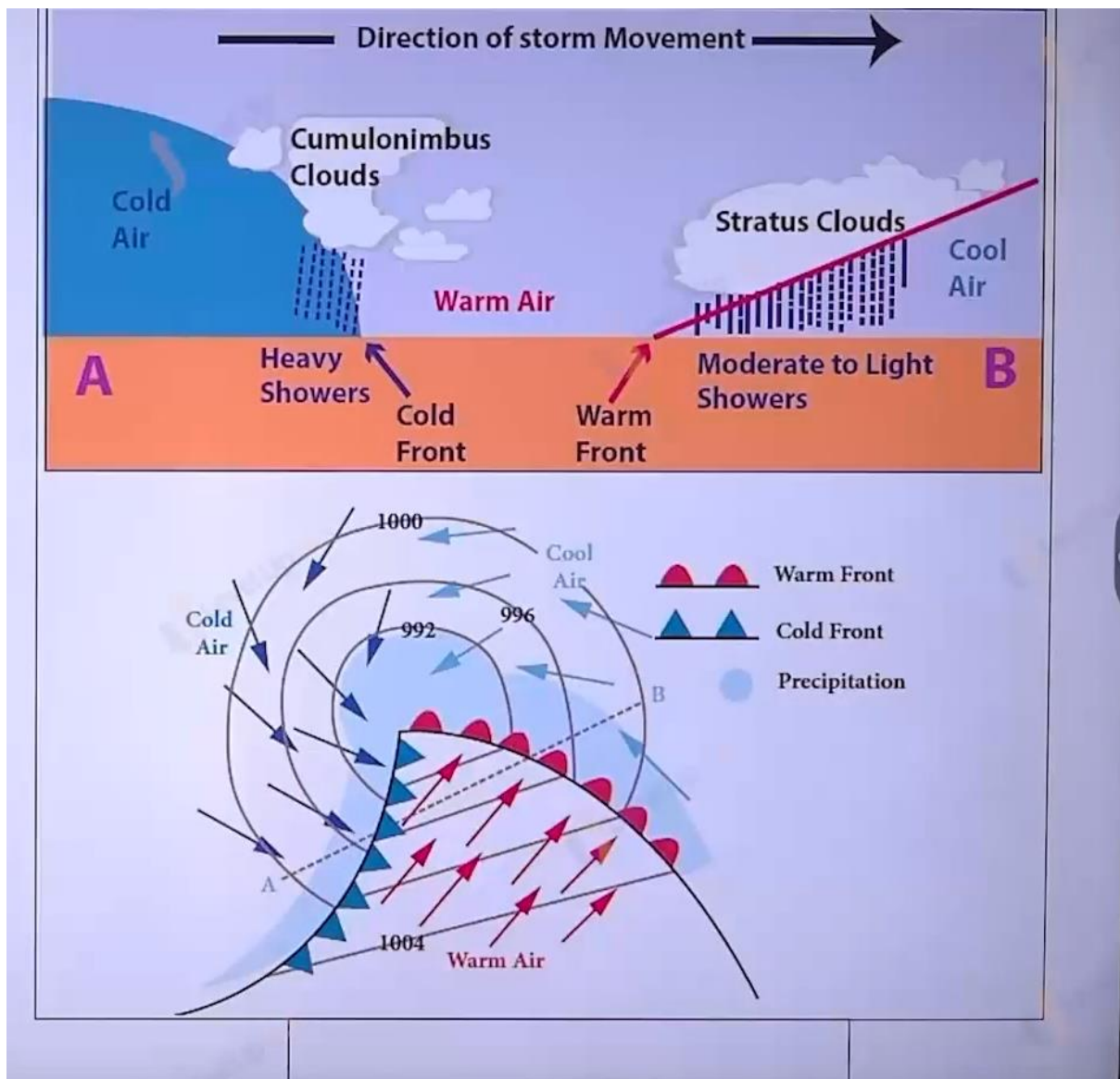
Structure of temperate cyclone

These cyclones are nothing but a rotating cold and warm fronts around a dynamically induced low pressure. All the weather phenomena are those which are associated with the cold and warm fronts.



Structure of temperate cyclone:

1. Temperate cyclones are nothing but the rotating system of cold front and warm front around the dynamically induced low pressure.
2. All the weather phenomena which are associated with the cold and warm front are experienced in the temperate cyclone.
- 3.



Tropical Cyclone

- Tropical cyclones are violent low pressure system that originate over oceans in tropical areas and move over to the coastal areas bringing about large scale destruction caused by violent winds, very heavy rainfall
- Move Anticlockwise in NH
- Move Clockwise in SH

How tropical storms are formed

High humidity and ocean temperatures of over 26°C are major contributing factors

Water evaporates from the ocean surface and comes into contact with a mass of cold air, forming clouds

A column of low pressure develops at the centre. Winds form around the column

As pressure in the central column (the eye) weakens, the speed of the wind around it increases



Favourable Conditions for the Formation of Tropical Cyclone

- A large area of water surface with a temperature above 27° C.
- The Coriolis force is strong enough to form a cyclonic vortex.
- A weak low-pressure area or low-level cyclonic circulation already exists on ocean surface
- There should be a divergence around 9 km to 15 km which sucks the air from ocean surface above and thus the upward movement of air is accelerated and low pressure centre at the surface is further intensified.
- Variations in the vertical wind speed are minor. This is referred to condition of **Low Wind Shear**
- Tropical cyclones develop around inter-tropical convergence zone

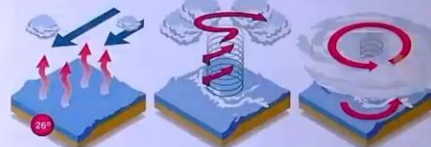
How tropical storms are formed

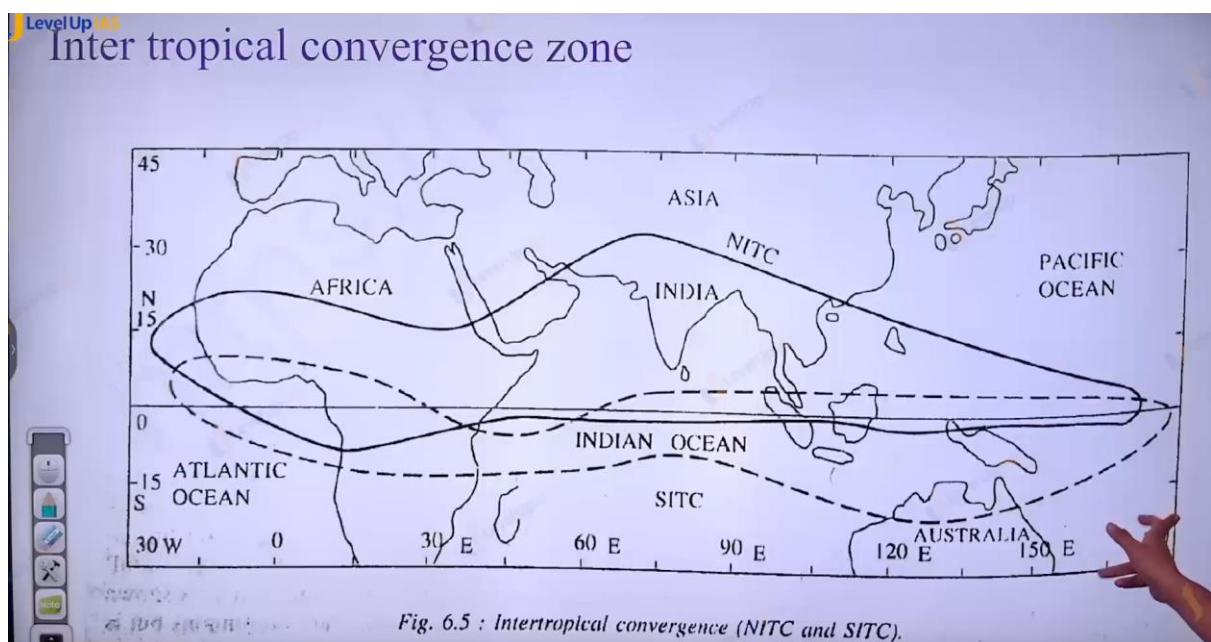
High humidity and ocean temperatures of over 26°C are major contributing factors

Water evaporates from the ocean surface and comes into contact with a mass of cold air, forming clouds

A column of low pressure develops at the centre. Winds form around the column

As pressure in the central column (the eye) weakens, the speed of the wind around it increases





Developmental Stages of Tropical Cyclone

Tropical cyclone development can be classified into 3 stages

1. Formation and Initial Development Stage

- Transport of water vapor and heat from the warm ocean to overlying air, largely through evaporation from the sea surface, leads to the creation of a cyclonic storm.

Condensation of rising air above the ocean surface stimulates the creation of huge vertical cumulus clouds.

Condensation releases heat and keeps



Tropical Cyclone:

1. Tropical cyclone are violent low pressure system that originates over the ocean in the tropical area. and moved to costal area and bring about large scale distribution due to volent ring and very heavy rain fall it move anti clockwise in northern hemisphere and clockwise in southern hemisphere.
- 2.

LevelUp IAS
2020

Consider the following statements:

1. Jet streams occur in the Northern Hemisphere **only**.
2. Only some cyclones develop an eye.
3. The temperature inside the eye of a cyclone is nearly 10 C lesser than that of the surroundings.

Which of the statements given above is/are correct:

(a) 1 only (b) 2 and 3 only

only (d) 1 and 3 only

LevelUp IAS
Mains Questions

- Q) Briefly mention the alignment of major mountain ranges of the world and explain their impact on local weather conditions, with examples. 2021
- Q) How do the melting of the Arctic ice and glaciers of the Antarctic differently affect the weather patterns and human activities on the Earth? Explain. 2021
- Q) How does the cryosphere affect global climate? 2017
- Q) Discuss the concept of air mass and explain its role in macro-climatic changes. 2016
- Q) Tropical cyclones are largely confined to the South China Sea, Bay of Bengal, and the Gulf of Mexico. Why? 2014

1. Major mountain ranges Himalia , Andes Rockies alps Urals , great dividing range darkening berk.
2. Their impact on the local weather condition
 - a. Moisture
 - b. g

